

Zanzibar: Google's Consistent, Global Authorization System

Ruoming Pang, Ramón Cáceres, Mike Burrows, Zhifeng Chen,
Pratik Dave, Nathan Germer, Alexander Golynski, Kevin Graney,
Nina Kang, Lea Kissner*, Jeffrey L. Korn, Abhishek Parmar,
Christina D. Richards, Mengzhi Wang
Google, LLC; Humu, Inc.; Carbon, Inc.
{rpang, caceres}@google.com

Abstract

Determining whether online users are authorized to access digital objects is central to preserving privacy. This paper presents the design, implementation, and deployment of Zanzibar, a global system for storing and evaluating access control lists. Zanzibar provides a uniform data model and configuration language for expressing a wide range of access control policies from hundreds of client services at Google, including Calendar, Cloud, Drive, Maps, Photos, and YouTube. Its authorization decisions respect causal ordering of user actions and thus provide external consistency amid changes to access control lists and object contents. Zanzibar scales to trillions of access control lists and millions of authorization requests per second to support services used by billions of people. It has maintained 95th-percentile latency of less than 10 milliseconds and availability of greater than 99.999% over 3 years of production use.

1 Introduction

Many online interactions require authorization checks to confirm that a user has permission to carry out an operation on a digital object. For example, web-based photo storage services typically allow photo owners to share some photos with friends while keeping other photos private. Such a service must check whether a photo has been shared with a user before allowing that user to view the photo. Robust authorization checks are central to preserving online privacy.

This paper presents Zanzibar, a system for storing permissions and performing authorization checks based on the stored permissions. It is used by a wide array of services offered by Google, including Calendar, Cloud, Drive, Maps, Photos, and YouTube. Several of these services manage billions of objects on behalf of more than a billion users.

A unified authorization system offers important advantages over maintaining separate access control mechanisms for individual applications. First, it helps establish consistent

* Work done while at Google.